

PRACTICE SHEET

1. What is the equivalent definition of the function given by

$$f(x) = \begin{cases} 2x, & x \geq 0 \\ 0, & x < 0 \end{cases}?$$
 - (a) $f(x) = |x|$
 - (b) $f(x) = 2x$
 - (c) $f(x) = |x| + x$
 - (d) $f(x) = 2|x|$
2. If $f: \mathbb{R} \rightarrow \mathbb{R}^+$ such that $f(x) = (1/3)^x$, then what is the value of $f^{-1}(x)$?
 - (a) $(1/3)^x$
 - (b) 3^x
 - (c) $\log_{1/3} x$
 - (d) $\log_x (1/3)$
3. Consider the following statements:
 The function $f(x) = \text{greatest integer } \leq x, x \in \mathbb{R}$ is a continuous function
 1. All trigonometric functions are continuous on $\mathbb{R}$.
 2. Which of the statements given above is/are correct?
 - (a) 1 only
 - (b) 2 only
 - (c) Both 1 and 2
 - (d) Neither 1 nor 2
4. If $f(x) = \log \left(\frac{1+x}{1-x} \right)$, then what is $f \left(\frac{2x}{1-x^2} \right)$ equal to?
 - (a) $(f(x))^2$
 - (b) 1
 - (c) $2f(x)$
 - (d) $f \left(\frac{1-x}{1+x} \right)$
5. If $-x^2 + 3x + 4 > 0$, then which one of the following is correct?
 - (a) $x \in (-1, 4)$
 - (b) $x \in [-1, 4]$
 - (c) $x \in (\infty, -1) \cup (4, \infty)$
 - (d) $x \in (-\infty, -1] \cup [4, \infty)$
6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) = x|x|$; for each $x \in \mathbb{R}$, $\mathbb{R}$ being the set of real numbers. Which of the following is correct?
 - (a) f is onto but not onto
 - (b) f is onto but not one – one
 - (c) f is both one – one and onto
 - (d) f is neither one – one nor onto
7. A mapping $f: \mathbb{R} \rightarrow \mathbb{R}$ which is defined as $f(x) = \cos x; x \in \mathbb{R}$ is
 - (a) One – one only
 - (b) Onto only
 - (c) One – one onto
 - (d) Neither one – one nor onto
8. Consider the following statements:
 I. Every function has a primitive
 II. A primitive of a function is unique
 Which of the statements given above is/are correct?
 - (a) Only I
 - (b) Only II
 - (c) Both I nor II
 - (d) Neither I nor II
9. If $f: \mathbb{R} \rightarrow \mathbb{R}, g: \mathbb{R} \rightarrow \mathbb{R}$ and $g(x) = x + 3$ and $(f \circ g)(x) = (x+3)^2$, then what is the value of $f(-3)$?
 - (a) -9
 - (b) 0
 - (c) 9
 - (d) 3
10. Consider the function $f: \mathbb{R} \rightarrow \{0,1\}$ such that:

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$$
 Which one of the following is correct?
 - (a) The function is one-one into
 - (b) The function is many-one into
 - (c) The function is one-one onto
 - (d) The function is many-one onto

Directions (for next three): Each item under List I is associated with one or more items under List II.

List I (Function)

 - A. $\sin x$
 - B. $\cos x$
 - C. $\tan x$

List II (Property)

 - (i) Periodic function
 - (ii) Non-periodic function
 - (iii) Continuous at every point on $(-\infty, \infty)$
 - (iv) Discontinuous function
 - (v) Differentiable at every point on $(-\infty, \infty)$
 - (vi) Not differentiable at every point on $(-\infty, \infty)$
 - (vii) Has period π
 - (viii) Has period 2π
 - (ix) Increases on $\left(0, \frac{\pi}{2}\right)$
 - (x) Decreases on $\left(0, \frac{\pi}{2}\right)$
 - (xi) Increases on $\left(\frac{\pi}{2}, \pi\right)$
 - (xii) Decreases on $\left(\frac{\pi}{2}, \pi\right)$
11. A is associated with
 - (a) (i), (iii), (v), (viii), (ix), (xxii)
 - (b) (ii), (iv), (vi), (viii), (x), (xxi)
 - (c) (i), (iii), (v), (vii), (x), (xxi)
 - (d) None of these
12. B is associated with
 - (a) (ii), (iii), (v), (viii), (ix), (xxii)
 - (b) (i), (iii), (v), (viii), (x), (xxii)
 - (c) (i), (iii), (v), (viii), (ix), (xxii)
 - (d) None of these
13. C is associated with
 - (a) (i), (iv), (vi), (vii), (ix), (xxi)
 - (b) (ii), (iv), (vi), (viii), (ix)
 - (c) (i), (iv), (vi), (vii), (ix)
 - (d) None of these
14. A mapping $f: \mathbb{R} \rightarrow \mathbb{R}$ which is defined as $f(x) = \cos x; x \in \mathbb{R}$ is:
 - (a) Only one-one
 - (b) Only onto
 - (c) One-one onto
 - (d) Neither one-one nor onto
15. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) = x|x|$; for each $x \in \mathbb{R}$, $\mathbb{R}$ being the set of real numbers. Which one of the following is correct?
 - (a) f is one-one but not onto
 - (b) f is onto but not one-one
 - (c) f is both one-one and onto
 - (d) f is neither one-one nor onto

ANSWER KEY

1.	c	2.	c	3.	d	4.	c	5.	a	6.	c	7.	d	8.	c	9.	c	10.	d
11.	a	12.	b	13.	a	14.	d	15.	c										

Solutions

Sol.1. (c)

The given function is $f(x) = \begin{cases} 2x, & x \geq 0 \\ 0, & x < 0 \end{cases}$

The equation can be re-written as

$$f(x) = \begin{cases} x + x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Hence equivalent definition of given function is $f(x) = |x| + x$

Sol.2. (c)

Given function is $f(x) = \left(\frac{1}{3}\right)^x$

Let $f(x) = y$, so, $y = \left(\frac{1}{3}\right)^x$

Taking $\log_{1/3}$ on both sides

$$\Rightarrow x \times \log_{1/3} \left(\frac{1}{3}\right) = \log_{(1/3)} y$$

$$x = \log_{(1/3)} y \\ \Rightarrow f^{-1}(x) = \log_{(1/3)} x$$

Sol.3. (d)

Here, greatest integer function $[x]$ is discontinuous at its integral value of x , $\cot x$ and $\operatorname{cosec} x$ are discontinuous at $0, \pi, 2\pi$ etc. and $\tan x$ and $\sec x$ are discontinuous at $x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$

etc. Therefore the greatest integer function and all trigonometric functions are not continuous for $x \in \mathbb{R}$

$$f(x) = -x^2 \text{ when } x < 0$$

Therefore, neither (1) nor (2) are true

Sol.4. (c)

Given that

$$x(x) = \log \left(\frac{1+x}{1-x} \right)$$

$$\text{so, } f \left(\frac{2x}{1+x^2} \right) = \log \left(\frac{1 + \frac{2x}{1+x^2}}{1 - \frac{2x}{1+x^2}} \right)$$

$$= \log \left(\frac{1+x^2+2x}{1+x^2-2x} \right) = \log \left(\frac{(1+x)^2}{(1-x)^2} \right)$$

$$= \log \left(\frac{1+x}{1-x} \right)^2 = 2 \log \left(\frac{1+x}{1-x} \right)$$

$$= 2f \left[f(x) \right] = \log \left(\frac{1+x}{1-x} \right)$$

Sol.5. (a)

$$-x^2 + 3x + 4 > 0$$

$$\Rightarrow x^2 - 3x - 4 < 0 \Rightarrow 0 < (x-4)(x+1) < 0$$

$$\Rightarrow x \in (-1, 4)$$

Sol.6. (c)

Given $f(x) = x|x|$

If $f(x_1) = f(x_2)$

$$\Rightarrow x_1|x_1| = x_2|x_2|$$

$$\Rightarrow x_1 : x_2$$

$\therefore f(x)$ is one - one.

Also, range of $f(x) = \mathbb{R}$ - domain of $f(x)$

$\therefore f(x)$ is onto.

Hence, $f(x)$ is both one - one and onto.

Sol.7. (d)

Let $x_1, x_2 \in \mathbb{R}$

Then, $f(x_1) = f(x_2)$

$$\Rightarrow \cos x_1 = \cos x_2$$

$$\Rightarrow x_1 = 2n\pi \pm x_2$$

$$\text{So, } x_1 \neq x_2$$

Hence, $\cos x$ is not one - one function.

Now, let $y = \cos x$

We know, $-1 \leq \cos x \leq 1$

$$\therefore y \in [-1, 1]$$

$[-1, 1] \subset \mathbb{R}$, so, $\cos x$ is into function, not onto.

Hence, $f(x) = \cos x$ is neither one - one nor onto.

Sol.8. (c)

primitive means preimage of elements of range.

Sol.9. (c)

$$g(x) = x + 3 \text{ and } f(g(x)) = (x + 3)^2$$

then it is clear that $f(x) = x^2$

$$f(-3) = (-3)^2 = 9$$

Sol.10. (d)

many one onto

Sol.11. (a)

by graph

Sol.12. (b)

by graph

Sol.13. (a)

by graph

Sol.14. (d)

Sol.15. (c)

$$f(x) = x^2 \text{ when } x > 0$$

NDA PYQ

1. If $f(xy) = f(x)f(y)$, then $f(t)$ may be of the form:
 (a) $t + k$ (b) $ct + k$
 (c) $t^k + c$ (d) t^k
[NDA (I) - 2012]
 2. Which of the following statements is correct?
 (a) e^x is an increasing function
 (b) e^x is a decreasing function
 (c) e^x is neither increasing nor decreasing function
 (d) e^x is a constant function
[NDA (I) - 2012]
 3. If $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function whose inverse is $\frac{x+5}{3}$, then what is the value of $f(x)$?
 (a) $f(x) = 3x + 5$ (b) $f(x) = 3x - 5$
 (c) $f(x) = 5x - 3$ (d) Does not exist
[NDA (II) - 2012]
 4. Consider the following statement:
 1. If $f(x) = x^3$ and $g(y) = y^3$ then $f = g$
 2. Identity function is not always a bijective
 Which of the following statements is/are true
 (a) 1 only (b) 2 only
 (c) both 1 and 2 (d) Neither 1 nor 2
[NDA (II) - 2012]
 5. Let $A = \{x \in \mathbb{R} \mid x \geq 0\}$ A function $f: A \rightarrow A$ is defined by $f(x) = x^2$, which of the following is correct?
 (a) The function does not have inverse
 (b) f is its own inverse
 (c) The function has an inverse but f is not its own inverse
 (d) None of above
[NDA (II) - 2012]
 6. What is the range of the function $f(x) = \frac{|x|}{x}$, where $x \neq 0$?
 (a) Set of all real numbers
 (b) Set of all integers
 (c) $\{-1, 1\}$
 (d) $\{-1, 0, 1\}$
[NDA (I) - 2013]
 7. Let N be the set of natural numbers and $f: N \rightarrow N$ be a function given by $f(x) = x+1$ for $x \in N$. Which one of the following is correct?
 (a) f is one-one and onto
 (b) f is one-one but not onto
 (c) f is only onto
 (d) f is neither one-one nor onto
[NDA (I) - 2013]
 8. If f be a function from the set of natural numbers to the set of even natural numbers given by $f(x) = 2x$. Then, f is:
 (a) One-one but not onto
 (b) Onto but not one-one
 (c) Both one-one and onto
 (d) Neither one-one nor onto
[NDA (II) - 2013]
 9. Let N denote the set of all non-negative integers and Z denote the set of all integers. The function $f: Z \rightarrow N$ given by $f(x) = |x|$ is:
 (a) One-one but not onto
 (b) Onto but not one-one
 (c) Both one-one and onto
 (c) Neither one-one nor onto
 - (c) Neither one-one nor onto
[NDA (I) - 2014]
 10. Consider the following relations from A to B where $A = \{u, v, w, x, y, z\}$ and $B = \{p, q, r, s\}$
 1. $\{(u,p), (v,p), (w,p), (x,q), (y,q), (z,q)\}$
 2. $\{(u,p), (v,q), (w,r), (z,s)\}$
 3. $\{(u,w), (v,r), (w,q), (u,p), (v,q), (z,q)\}$
 4. $\{(u,q), (v,p), (w,s), (x,r), (y,q), (z,s)\}$
 Which of the above relations are not functions?
 (a) 1 and 2 (b) 1 and 4
 (c) 2 and 3 (d) 3 and 4
[NDA-2014(I)]
 11. Consider the following statements:
 1. The function $f(x) = \sin x$ decreases on the interval $(0, \pi/2)$
 2. The function $f(x) = \cos x$ increases on the interval $(0, \pi/2)$
 Which of the above statements is/are correct?
 (a) 1 only (b) 2 only
 (c) both 1 and 2 (d) neither 1 nor 2
[NDA-2014(I)]
 12. The function $f: N \rightarrow N$, where N being the set of natural numbers, defined by $f(x) = 2x + 3$ is:
 (a) Injective and surjective
 (b) Injective but not surjective
 (c) Not injective but surjective
 (d) Neither injective nor surjective
[NDA (II) - 2014]
- Direction (for next three):** Read the following information carefully and answer the questions given below:
- Consider the function $f(x) = \frac{x-1}{x+1}$
13. What is $\frac{f(x)+1}{f(x)-1} + x$ equal to?
 (a) 0 (b) 1
 (c) $2x$ (d) $4x$
[NDA (II) - 2014]
 14. What is $f(2x)$ equal to?
 (a) $\frac{f(x)+1}{f(x)+3}$ (b) $\frac{f(x)+1}{3f(x)+1}$
 (c) $\frac{3f(x)+1}{f(x)+3}$ (d) $\frac{f(x)+3}{3f(x)+1}$
[NDA (II) - 2014]
 15. What is $f[f(x)]$ equal to?
 (a) x (b) $-x$
 (c) $-\frac{1}{x}$ (d) None of these
[NDA (II) - 2014]
 16. Let $f(x) = ax + b$ and $g(x) = cx + d$. Then $f(g(x)) = g(f(x))$ is equivalent to:
 (a) $f(c) = g(a)$ (b) $f(a) = g(c)$
 (c) $f(c) = g(d)$ (d) $f(d) = g(b)$
[NDA (II) - 2014]
 17. Consider the following functions:
 I. $f(x) = x^3, x \in \mathbb{R}$
 II. $f(x) = \sin x, 0 < x < 2\pi$
 III. $f(x) = e^x, x \in \mathbb{R}$
 Which of the above functions have inverse defined on their ranges?

- (a) I and II
(c) I and III
- (b) II and III
(d) I, II and III

[NDA (I) - 2015]

18. If $f(x) = \log_e \left(\frac{1+x}{1-x} \right)$, $g(x) = \frac{3x+x^3}{1+3x^2}$ and $\text{gof}(t) = g(f(t))$, then what is $\text{gof} \left(\frac{e-1}{e+1} \right)$ equal to?
- (a) 2
(c) 0
- (b) 1
(d) 1/2

[NDA (I) - 2015]

19. For each non zero real number x , let $f(x) = \frac{x}{|x|}$, the range of function is
- (a) a null set
(b) a set consisting of only one element
(c) a set consisting of two element
(d) a set consisting of infinitely many elements

[NDA (I) - 2015]

20. If $g(x) = \frac{1}{f(x)}$ and $f(x) = x$, $x \neq 0$, then which one of the following is correct?
- (a) $f(f(f(g(f(x)))))) = g(g(f(g(f(x))))))$
(b) $f(f(g(g(f(x)))))) = g(g(f(g(f(x))))))$
(c) $f(g(g(f(g(f(x)))))) = g(g(f(g(f(x))))))$
(d) $f(f(f(g(f(x)))))) = f(f(f(g(f(x))))))$

[NDA (II) - 2015]

21. The domain of the function $f(x) = \frac{1}{\sqrt{|x|-x}}$ is
- (a) $[0, \infty)$
(c) $[1, \infty)$
- (b) $(-\infty, 0)$
(d) $(-\infty, 0]$

[NDA (II) - 2015]

22. $f(xy) = f(x) + f(y)$ is true for all
- (a) Polynomial function
(c) Exponential functions
- (b) Trigonometric functions
(d) Logarithmic functions

[NDA (II) - 2015]

23. Consider the following statements.
Statement I: The function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = x^3$ for all $x \in \mathbb{R}$ is one-one.
Statement II: $f(a) = f(b) \Rightarrow a = b$ for all $a, b \in \mathbb{R}$ if the function f is one-one
Which one of the following is correct in respect of the above statements?
- (a) Both the statements are true and statement II is the correct explanation of statement I.
(b) Both the statements are true and statement II is not the correct explanation of statement I.
(c) Statement I is true but statement II is false
(d) Statement I is false but statement II is true.

[NDA (II) - 2015]

24. If $f: \mathbb{R} \rightarrow \mathbb{R}$, $g: \mathbb{R} \rightarrow \mathbb{R}$ be two functions given by $f(x) = 2x - 3$ and $g(x) = x^3 + 5$, then $(\text{fog})^{-1}(x)$ is equal to:
- (a) $\left(\frac{x+7}{2} \right)^{\frac{1}{3}}$
(c) $\left(x - \frac{7}{2} \right)^{\frac{1}{3}}$
- (b) $\left(\frac{x-7}{2} \right)^{\frac{1}{3}}$
(d) $\left(x + \frac{7}{2} \right)^{\frac{1}{3}}$

[NDA (II) - 2015]

25. What is the range of the function $y = \frac{x^2}{1+x^2}$ where $x \in \mathbb{R}$?

- (a) $[0, 1)$
(c) $(0, 1)$
- (b) $[0, 1]$
(d) $(0, 1]$

[NDA (I) - 2016]

26. If $f(x_1) - f(x_2) = f \left(\frac{x_1 - x_2}{1 - x_1 x_2} \right)$ for, $x_1, x_2 \in (-1, 1)$, then what is $f(x)$ equal to?
- (a) $\ln \left(\frac{1-x}{1+x} \right)$
(c) $\tan^{-1} \left(\frac{1-x}{1+x} \right)$
- (b) $\ln \left(\frac{2+x}{1-x} \right)$
(d) $\tan^{-1} \left(\frac{1+x}{1-x} \right)$

[NDA (I) - 2016]

Direction (for next two): Let $f(x)$ be the greatest integer function and $g(x)$ be the modulus function.

27. What is the value of $(\text{gof}) \left(-\frac{5}{3} \right) - (\text{fog}) \left(-\frac{5}{3} \right)$
- (a) -1
(c) 1
- (b) 0
(d) 2

[NDA (I) - 2016]

28. What is the value of $(\text{fog}) \left(-\frac{9}{5} \right) + (\text{gog})(-2)$
- (a) -1
(c) 1
- (b) 0
(d) 2

[NDA (I) - 2016]

29. What is the domain of the function $f(x) = \frac{1}{\sqrt{|x|-x}}$?
- (a) $(-\infty, 0)$
(c) $0 < x < 1$
- (b) $(0, \infty)$
(d) $x > 1$

[NDA (II) - 2016]

30. If $f(x) = \frac{x}{x-1}$, then what is $\frac{f(a)}{f(a+1)}$ equal to:
- (a) $f \left(-\frac{a}{a+1} \right)$
(c) $f \left(\frac{1}{a} \right)$
- (b) $f(a^2)$
(d) $f(-a)$

[NDA (I) - 2017]

31. Let $f(a) = \frac{a-1}{a+1}$, Consider the following:

1. $f(2a) = f(a) + 1$ 2. $f \left(\frac{1}{a} \right) = -f(a)$

Which of the above is / are correct?

- (a) Only 1
(c) Both 1 and 2
- (b) Only 2
(d) neither 1 nor 2

[NDA (I) - 2017]

32. The function $f: X \rightarrow Y$ defined by $f(x) = \cos x$, where $x \in X$, is one-one and onto if X and Y are respectively equal to:
- (a) $[0, \pi]$ and $[-1, 1]$
(b) $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$ and $[-1, 1]$
(c) $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$ and $[0, 1]$
(d) $[0, \pi]$ and $[0, 1]$

[NDA (I) - 2017]

33. Let $f(x) = px + q$ and $g(x) = mx + n$. Then $f(g(x)) = g(f(x))$ is equivalent to:
- (a) $f(p) = g(m)$
(b) $f(q) = g(n)$

(c) $f(n) = g(q)$

(d) $f(m) = g(p)$

[NDA (I) - 2017]

34. Let $f: [-6, 6] \rightarrow \mathbb{R}$ be defined by $f(x) = x^2 - 3$. Consider the following:

I. $(f \circ f \circ f)(-1) = (f \circ f \circ f)(1)$

II. $(f \circ f \circ f)(-1) - 4 = (f \circ f \circ f)(1) = (f \circ f)(0)$

Which of the above is/are correct?

(a) Only I

(b) Only II

(c) Both I and II

(d) Neither I nor II

[NDA (I) - 2017]

35. Let $f(x) : \begin{cases} x, & x \text{ is rational} \\ 0, & x \text{ is irrational} \end{cases}$
and $g(x) : \begin{cases} 0, & x \text{ is rational} \\ x, & x \text{ is irrational} \end{cases}$

If $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$, then $(f \circ g)$ is:

(a) one-one and into

(b) neither one-one nor onto

(c) many-one and onto

(d) one-one and onto

[NDA-2017(1)]

36. Which one of the following functions is neither even nor odd?

(a) $x^2 - 1$

(b) $x + 3/x$

(c) $|x|$

(d) $x^2(x-3)$

[NDA-2017(1)]

37. The function $f(x) = |x| - x^3$ is:

(a) Odd

(b) Even

(c) Both even and odd

(d) Neither even nor odd

[NDA (II) - 2017]

38. If x is any real number, then $\frac{x^2}{1+x^4}$ belongs to which one of the following intervals?

(a) $(0, 1)$

(b) $\left(0, \frac{1}{2}\right]$

(c) $\left[0, \frac{1}{2}\right]$

(d) $[0, 1]$

[NDA (II) - 2017]

39. The inverse of the function $y = 5^{\ln x}$ is:

(a) $x = y^{\frac{1}{\ln 5}}, y > 0$

(b) $x = y^{\ln 5}, y > 0$

(c) $x = y^{\frac{1}{\ln 5}}, y < 0$

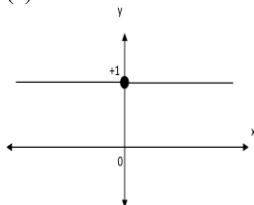
(d) $x = y^{\ln y}, y > 0$

[NDA (II) - 2017]

40. Which of the following graphs represents the function

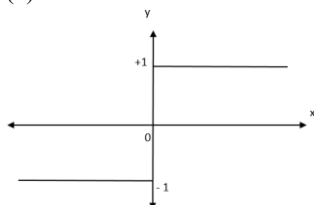
$f(x) = \frac{x}{x}, x \neq 0$

(a)

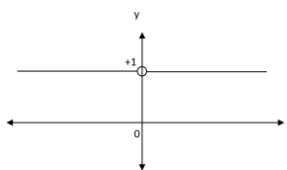


(c)

(b)



(d) None of above



[NDA (II) - 2017]

41. Let $f(n) = \left[\frac{1}{4} + \frac{n}{1000} \right]$, where $[x]$ denotes the greatest integer

function. Then the value of $\sum_{n=1}^{1000} f(n)$ is

(a) 251

(b) 250

(c) 1

(d) 0

[NDA (II) - 2017]

42. If $f(x) = \frac{4x+x^4}{1+4x^3}$ and $g(x) = \ln \left(\frac{1+x}{1-x} \right)$, then what is the value of $\text{fog} \left(\frac{e-1}{e+1} \right)$ equal to:

(a) 2

(b) 1

(c) 0

(d) $1/2$

[NDA (II) - 2017]

43. Let $[x]$ denote the greatest integer function. What is the number of solutions of the equation $x^2 - 4x + [x] = 0$ in the interval $[0, 2]$?

(a) Zero (no solution)

(b) One

(c) Two

(d) Three

[NDA (I) - 2018]

44. What is the period of the function $f(x) = \sin x$?

(a) $\pi/4$

(b) $\pi/2$

(c) π

(d) 2π

[NDA (I) - 2018]

45. For f to be a function, what is the domain of f , if $f(x) = \frac{1}{\sqrt{|x| - x}}$?

(a) $(-\infty, 0)$

(b) $(0, \infty)$

(c) $(-\infty, \infty)$

(d) $(-\infty, 0]$

[NDA (I) - 2018]

46. If $f: \mathbb{R} \rightarrow \mathbb{S}$ defined by $f(x) = 4 \sin x - 3 \cos x + 1$ is onto, then what is $\mathbb{S}$ equal to?

(a) $[-5, 5]$

(b) $(-5, 5)$

(c) $(-4, 6)$

(d) $[-4, 6]$

[NDA (I) - 2018]

47. Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = \frac{x^2}{1+x^2}$. What is the range of the function?

(a) $[0, 1)$

(b) $[0, 1]$

(c) $(0, 1]$

(d) $(0, 1)$

[NDA (I) - 2018]

48. Which one of the following is correct in respect of the function $f: \mathbb{R} \rightarrow \mathbb{R}^+$ defined as $f(x) = |x+1|$?

(a) $f(x^2) = [f(x)]^2$

(b) $f(|x|) = |f(x)|$

(c) $f(x+y) = f(x) + f(y)$

(d) None of the above

[NDA (I) - 2018]

49. If $f(x)$ is an even function, where $f(x) \neq 0$, then which of the following is correct?

(a) $f(x)$ is even function

(b) $f(x)$ is odd function

(c) $f(x)$ may be an even or odd function depending on the type of function

(d) $f(x)$ is a constant function

[NDA (I) - 2018]

50. If $f(x) = \frac{\sqrt{x-1}}{x-4}$ defines a function of R , then what is domain?

- (a) $(-\infty, 4) \cup (4, \infty)$ (b) $(4, \infty)$
(c) $(1, 4) \cup (4, \infty)$ (d) $[1, 4) \cup (4, \infty)$

[NDA (II) - 2018]

51. A function $f: A \rightarrow R$ is defined by the equation $f(x) = x^2 - 4x + 5$ where $A = (1, 4)$. What is the range of the function?

- (a) $(2, 5)$ (b) $(1, 5)$
(c) $[1, 5)$ (d) $[1, 5]$

[NDA (II) - 2018]

52. Let $A = \{x \in R: -1 \leq x \leq 1\}$ and S be the subset of $A \times B$, defined by: $S = \{(x, y) \in A \times B: x^2 + y^2 = 1\}$

Which one of the following is correct?

- (a) S is a one-one function from A into B
(b) S is a many one function from A into B
(c) S is a objective mapping form A into B
(d) S is not a function

[NDA (II) - 2018]

53. If $f(x) = \frac{x-2}{x+2}$, $x \neq -2$ is then $f^{-1}(x)$ equal to?

- (a) $\frac{4(x+2)}{x-2}$ (b) $\frac{(x+2)}{4(x-2)}$
(c) $\frac{(x+2)}{x-2}$ (d) $\frac{2(x+1)}{1-x}$

[NDA (I) - 2019]

54. For $r > 0$, $f(r)$ is the ratio of perimeter to area of a circle of radius r . Then $f(1)+f(2)$ is equal to

- (a) 1 (b) 2
(c) 3 (d) 4

[NDA (I) - 2019]

55. If $f(x) = 3^{1+x}$, then $f(x) \cdot f(y) \cdot f(z)$ is equal to

- (a) $f(x+y+z)$ (b) $f(x+y+z+1)$
(c) $f(x+y+z+2)$ (d) $f(x+y+z+3)$

[NDA (I) - 2019]

56. The domain of the function $\sqrt{(2-x)(x-3)}$

- (a) $(0, \infty)$ (b) $[0, \infty)$
(c) $[2, 3]$ (d) $(2, 3)$

[NDA (I) - 2019]

57. A function f defined by $f(x) = \ln(\sqrt{x^2+1} - x)$ is

- (a) An even function
(b) An odd function
(c) Both even and odd function
(d) Neither even nor odd function

[NDA (I) - 2019]

Direction (for next three) :

Read the following information and answer the three items that follow:

Let $f(x) = x^2 + 2x - 5$ and $g(x) = 5x + 30$

58. What are the roots of the equation $g[f(x)] = 0$

- (a) 1, -1 (b) -1, -1
(c) 1, 1 (d) 0, 1

[NDA (II) - 2019]

59. Consider the following statements:

1. $f[g(x)]$ is a polynomial of degree 3
2. $g[g(x)]$ is a polynomial of degree 2
Which of the above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (II) - 2019]

60. If $h(x) = 5f(x) - xg(x)$, then what is the derivative of $h(x)$?

- (a) -40 (b) -20
(c) -10 (d) 0

[NDA (II) - 2019]

61. Which one of the following is the second degree polynomial function $f(x)$ where $f(0) = 5$, $f(-1) = 10$ and $f(1) = 6$?

- (a) $5x^2 - 2x + 5 = 0$ (b) $3x^2 - 2x - 5 = 0$
(c) $3x^2 - 2x + 5 = 0$ (d) $3x^2 - 10x + 5 = 0$

[NDA (II) - 2019]

Direction for next two:

Let $f(x) = x^2$, $g(x) = \tan x$ and $h(x) = \ln x$.

62. For $x = \frac{\sqrt{\pi}}{2}$, what is the value of $[ho(gof)](x)$?

- (a) 0 (b) 1
(c) $\pi/4$ (d) $\pi/2$

[NDA (II) - 2019]

63. What is $[fo(fof)](2) =$

- (a) 2 (b) 8
(c) 16 (d) 256

[NDA (II) - 2019]

Direction for next three:

Consider the function $f(x) = g(x) + h(x)$

where $g(x) = \sin\left(\frac{x}{4}\right)$ and $h(x) = \cos\left(\frac{4x}{5}\right)$

64. What is the period of function $g(x)$?

- (a) π (b) 2π
(c) 4π (d) 8π

[NDA (II) - 2019]

65. What is the period of function $h(x)$?

- (a) π (b) $4\pi/5$
(c) $5\pi/2$ (d) $3\pi/2$

[NDA (II) - 2019]

66. What is the period of function $f(x)$?

- (a) 10π (b) 20π
(c) 40π (d) 80π

[NDA (II) - 2019]

67. If $f(x) = 3x^2 - 5x + p$ and $f(0)$ and $f(1)$ are opposite sign, then which of the following is correct?

- (a) $-2 < p < 0$ (b) $-2 < p < 2$
(c) $0 < p < 2$ (d) $3 < p < 5$

[NDA 2020]

68. What is the domain of the function $f(x) = \cos^{-1}(x-2)$?

- (a) $[-1, 1]$ (b) $[1, 3]$
(c) $[0, 5]$ (d) $[-2, 1]$

[NDA 2020]

69. If $f(x) = 2x - x^2$, then what is the value of $f(x+2) + f(x-2)$ when $x = 0$?

- (a) -8 (b) -4
(c) 8 (d) 4

[NDA 2020]

70. Which one of the following is correct in respect of the graph of $y = \frac{1}{x-1}$?

- (a) The domain is $\{x \in R | x \neq 1\}$ and the range of the set of reals.
(b) The domain is $\{x \in R | x \neq 1\}$, the range is $\{y \in R | y \neq 0\}$, and the graph intersects y-axis at $(0, -1)$
(c) The domain is the set of reals and the range is singleton set $\{0\}$.
(d) The domain is $\{x \in R | x \neq 1\}$, the range is the set of points on the y axis.

71. Consider the following statements:
1. A function $f: Z \rightarrow Z$, defined by $f(x) = x + 1$, is one as well as onto.
2. A function $f: N \rightarrow N$, defined by $f(x) = x + 1$, is one but not onto.
Which of the above is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II [NDA 2020]
72. What is the domain of the function $f(x) = 3^x$
(a) $(-\infty, \infty)$ (b) $(0, \infty)$
(c) $[0, \infty)$ (d) $(-\infty, \infty) - \{0\}$ [NDA 2021 (I)]
73. If $f(x+1) = x^2 - 3x + 2$, then what is $f(x)$ equal to?
(a) $x^2 - 5x + 4$ (b) $x^2 - 5x + 6$
(c) $x^2 + 3x + 3$ (d) $x^2 - 3x + 1$ [NDA (I) - 2021]
74. What is the period of the function $f(x) = \ln(2 + \sin^2 x)$
(a) $\frac{\pi}{2}$ (b) π
(c) 2π (d) 3π [NDA (I) - 2021]
75. What is the range of the function $f(x) = 1 - \sin x$ defined on entire real line?
(a) $(0, 2)$ (b) $[0, 2]$
(c) $(-1, 1)$ (d) $[-1, 1]$ [NDA (II) - 2021]
76. Consider the following statements in respect of relations and functions:
1. all relations are functions but all functions are not relation.
2. A relation from A to B is a subset of Cartesian product $A \times B$
3. A relation in A is a subset of Cartesian product $A \times A$
Which of the above statements are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3 [NDA (II) 2021]
77. If $4f(x) - f\left(\frac{1}{x}\right) = \left(2x + \frac{1}{x}\right)\left(2x - \frac{1}{x}\right)$, then what is $f(2)$ equal to?
(a) 0 (b) 1
(c) 2 (d) 4 [NDA (I) - 2022]
78. If $f(x) = 4x + 3$, then what is $f \circ f \circ f(-1)$ equal to?
(a) -1 (b) 0
(c) 1 (d) 2 [NDA (I) - 2022]
79. Consider the following in respect of the function $f(x) = 10^x$.
1. Its domain is $(-\infty, \infty)$
2. It is a continuous function
3. It is differentiable at $x = 0$
Which of the above statement are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3 [NDA (I) - 2022]
80. What is the domain of the function $f(x) = \sqrt{1 - (x-1)^2}$?
(a) $(0, 1)$ (b) $[-1, 1]$
(c) $(0, 2)$ (d) $[0, 2]$ [NDA (I) - 2022]
81. Let $A = \{7, 8, 9, 10, 11, 12, 13, 14, 15, 16\}$ and let $f: A \rightarrow N$ be defined by $f(x) =$ the highest prime factor of x . How many elements are there in the range of f ?
(a) 4 (b) 5 [NDA 2020]
- (c) 6 (d) 7 [NDA 2022 (II)]
82. Let $z = [y]$ and $y = [x] - x$, where $[.]$ is the greatest integer function. If x is not an integer but positive, then what is the value of z ?
(a) -1 (b) 0
(c) 1 (d) 2 [NDA 2022 (II)]
83. If $f(x) = 4x + 1$ and $g(x) = kx + 2$ such the $\text{fog}(x) = \text{gof}(x)$, then what is the value of k ?
(a) 7 (b) 5
(c) 4 (d) 3 [NDA 2022 (II)]
84. If $f(\alpha) = \sqrt{\sec^2 \alpha - 1}$, then what is $\frac{f(\alpha) + f(\beta)}{1 - f(\beta)f(\alpha)}$ equal to?
(a) $f(\alpha - \beta)$ (b) $f(\alpha + \beta)$
(c) $f(\alpha)f(\beta)$ (d) $f(\alpha\beta)$ [NDA 2022 (II)]
85. If $f(x) = \ln\left(x + \sqrt{1 + x^2}\right)$, then which one of the following is correct?
(a) $f(x) + f(-x) = 0$ (b) $f(x) - f(-x) = 0$
(c) $2f(x) = f(-x)$ (d) $f(x) = 2f(-x)$ [NDA 2022 (II)]
86. Consider the following statements:
1. If f is the subset of $X \times Z$ defined by $f = \{(xy, x-y); x, y \in Z\}$, then f is a function from Z to Z .
2. If f is the subset of $N \times N$ defined by $f = \{(xy, x+y); x, y \in N\}$, then f is a function from N to N .
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2 [NDA 2022 (II)]
87. If $f(x) = x^2 + 2$ and $g(x) = 2x - 3$, then what is $(fg)(1)$ equal to?
(a) 3 (b) 1
(c) -2 (d) -3 [NDA - 2023 (I)]
88. What is the range of the function $f(x) = x + |x|$ if the domain is the set of real numbers?
(a) $(0, \infty)$ (b) $[0, \infty)$
(c) $(-\infty, \infty)$ (d) $[1, \infty)$ [NDA - 2023 (I)]
89. If $f(x) = x(4x^2 - 3)$, then what is $f(\sin \theta)$ equal to?
(a) $-\sin 3\theta$ (b) $-\cos 3\theta$
(c) $\sin 3\theta$ (d) $-\sin 4\theta$ [NDA - 2023 (I)]
- Consider the following for the next two (02) items that follow:
Let $f(x) = \sin [\pi^2] x + \cos [-\pi^2] x$ where $[.]$ is a greatest integer function.
90. What is $f\left(\frac{\pi}{2}\right)$ equal to?
(a) -1 (b) 0
(c) 1 (d) 2 [NDA - 2023 (I)]
91. What is $f\left(\frac{\pi}{4}\right)$ equal to?
(a) $-\frac{1}{\sqrt{2}}$ (b) -1

(c) 1

(d) $\frac{1}{\sqrt{2}}$

[NDA – 2023 (1)]

92. A mapping $f:A \rightarrow B$ defined as $f(x) = \frac{2x+3}{3x+5}$, $x \in A$. If f is to be onto, then what are A and B equal to?

(a) $A = \mathbb{R} \setminus \left\{-\frac{5}{3}\right\}$ and $B = \mathbb{R} \setminus \left\{-\frac{2}{3}\right\}$

(b) $A = \mathbb{R}$ and $B = \mathbb{R} \setminus \left\{-\frac{5}{3}\right\}$

(c) $A = \mathbb{R} \setminus \left\{-\frac{3}{2}\right\}$ and $B = \mathbb{R} \setminus (0)$

(d) $A = \mathbb{R} \setminus \left\{-\frac{5}{3}\right\}$ and $B = \mathbb{R} \setminus \left\{\frac{2}{3}\right\}$

[NDA – 2023 (1)]

93. Consider the following statements:

1. The relation f defined by

$$f(x) = \begin{cases} x^3, & 0 \leq x \leq 2 \\ 4x, & 2 \leq x \leq 8 \end{cases} \text{ is a function}$$

2. The relation g defined by

$$g(x) = \begin{cases} x^2, & 0 \leq x \leq 4 \\ 3x, & 4 \leq x \leq 8 \end{cases} \text{ is a function}$$

which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA-2023 (2)]

94. A function satisfies $f(x-y) = \frac{f(x)}{f(y)}$, where $f(y) \neq 0$. If $f(1)=0.5$, then what is $f(2) + f(3) + f(4) + f(5) + f(6)$ equal to?

- (a) 15/32 (b) 17/32
(c) 29/64 (d) 31/64

[NDA-2023 (2)]

Consider the following for the next (02) items that follow

Let $f(x) = x^2 - 1$ and $\text{gof}(x) = x - \sqrt{x} + 1$

95. Which one of the following is a possible expression for $g(x)$?

- (a) $\sqrt{x+1} - \sqrt[4]{x+1}$ (b) $\sqrt{x+1} - \sqrt[4]{x+1} + 1$
(c) $\sqrt{x+1} + \sqrt[4]{x+1}$ (d) $x+1 - \sqrt{x+1} + 1$

[NDA-2023 (2)]

96. What is $g(15)$ equal to?

- (a) 1 (b) 2
(c) 3 (d) 4

[NDA-2023 (2)]

Consider the following for the next (02) items that follow

Let a function f be defined on $\mathbb{R} - (0)$ and $2f(x) + f(1/x) = x + 3$.

97. What is $f(0.5)$ equal to?

- (a) 1/2 (b) 2/3
(c) 1 (d) 2

[NDA-2023 (2)]

98. If f is differentiable, then what is $f'(0.5)$ equal to?

- (a) 1/4 (b) 2/3
(c) 2 (d) 4

[NDA-2023 (2)]

Consider the following for the next (02) items that follow

A function is defined by $f(x) = \pi + \sin^2 x$

99. What is the range of the function?

- (a) $[0, 1]$ (b) $[\pi, \pi+1]$
(c) $[\pi-1, \pi+1]$ (d) $[\pi-1, \pi-1]$

[NDA-2023 (2)]

100. What is the period of the function?

- (a) 2π
(b) π
(c) $\pi/2$
(d) the function is non-periodic

[NDA-2023 (2)]

101. What is the domain of the function $f(x) = \sqrt{2-x} + \sqrt{2+x}$?

- (a) $(-2, 2)$ (b) $[-2, 2]$
(c) $\mathbb{R} - (-2, 2)$ (d) $\mathbb{R} - [-2, 2]$

[NDA-2023 (2)]

Consider the following for the next item that follow:

$$\text{Let } 3f(x) + f\left(\frac{1}{x}\right) = \frac{1}{x} + 1$$

102. What is $f(x)$ equal to?

- (a) $\frac{1}{8x} - \frac{x}{8} + \frac{1}{4}$ (b) $\frac{3}{8x} - \frac{x}{8} + \frac{3}{4}$
(c) $\frac{3}{8x} + \frac{x}{8} + \frac{1}{4}$ (d) $\frac{3}{8x} - \frac{x}{8} + \frac{1}{4}$

[NDA-2024 (1)]

103. If $f(x) = ax - b$ and $g(x) = cx + d$ are such that $f(g(x)) = g(f(x))$, then which one of the following holds?

- (a) $f(d) = g(b)$ (b) $f(b) + g(d) = 0$
(c) $f(d) + g(c) = 2a$ (d) $f(d) + g(b) = 2d$

[NDA-2024 (1)]

104. Which one of the following is correct in respect of

$$f(x) = \frac{1}{\sqrt{|x| - x}} \text{ and } g(x) = \frac{1}{\sqrt{x - |x|}}?$$

- (a) $f(x)$ has some domain and $g(x)$ has no domain
(b) $f(x)$ has no domain and $g(x)$ has some domain
(c) $f(x)$ and $g(x)$ have the same domain
(d) $f(x)$ and $g(x)$ do not have any domain

[NDA-2024 (1)]

105. Let $A = \{x \in \mathbb{R} : -1 < x < 1\}$. Which of the following is/are bijective functions from A to itself?

1. $f(x) = x|x|$
2. $g(x) = \cos(\pi x)$

Select the correct answer using the code given below:

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA-2024 (1)]

106. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{6, 7\}$. What is the number of onto functions from A to B ?

- (a) 10 (b) 20
(c) 30 (d) 32

[NDA-2024 (1)]

Direction: Consider the following for the one items given below

$$\text{Let } f(t) = \ln\left(t + \sqrt{1+t^2}\right) \text{ and } g(t) = \tan(f(t))$$

107. Consider the following statements:

I. $f(t)$ is an odd function.

II. $g(t)$ is an odd function.

Which of the statements given above is/are correct?

- (a) I only (b) II only
(c) Both I and II (d) Neither I nor II

[NDA-2024 (2)]

Direction: Consider the following for the **two items** given below

Let $f \circ g(x) = \cos^2 \sqrt{x}$ and $g \circ f(x) = |\cos x|$

108. Which one of the following is $f(x)$?

- (a) $\cos x$ (b) $\cos x^2$
(c) $\cos^2 x$ (d) $\cos |x|$

[NDA-2024 (2)]

109. Which one of the following is $g(x)$?

- (a) $\sqrt{x}$ (b) $|x|$
(c) x^2 (d) $x|x|$

[NDA-2024 (2)]

Direction: Consider the following for the **one items** given below

Let $f(x) = [x]^2 - [x^2]$

110. What is $f(0.999) + f(1.001)$

- (a) -1 (b) 0
(c) 1 (d) 2

[NDA-2024 (2)]

111. Let $z = [y]$ and $y = [x] - x$, where $[.]$ is the greatest integer function. If x is not an integer but positive, then what is the value of z ?

- (a) -1 (b) 0
(c) 1 (d) 2

[NDA-2024 (2)]

112. If $f(x) = 4x + 1$ and $g(x) = kx + 2$ such that $f \circ g(x) = g \circ f(x)$, then what is the value of k ?

- (a) 7 (b) 5
(c) 4 (d) 3

[NDA-2024 (2)]

113. If $f(x) = 4x^2 + 1$, then for how many real values of x will $f(2x)$ be the GM of $f(x)$ and $f(4x)$?

- (a) Four (b) Two
(c) One (d) none

[NDA-2024 (2)]

114. If $f(x) = [x]^2 - 30[x] + 221 = 0$, where $[x]$ is the greatest integer function, then what is the sum of all integer solutions?

- (a) 13 (b) 17
(c) 27 (d) 30

[NDA-2024 (2)]

115. Let $f(x)f(y) = f(xy)$ for all real x, y . If $f(2) = 4$, then what is the value of $f(1/2)$?

- (a) $1/4$ (b) $1/2$
(c) 1 (d) 4

[NDA-2024 (2)]

Consider the following for the one (01) items that follow:

116. Let the function $y = (1 - \cos x)^{-1}$, where $x \neq 2n\pi$ and n is an integer.

What is the range of the function?

- (a) $[0, \infty)$ (b) $[0.5, \infty)$
(c) $[1, \infty)$ (d) $(-\infty, 0.5]$

[NDA-2025 (1)]

Consider the following for the one (01) items that follow:

Let the curve $f(x) = |x-3|$

117. What is the domain of the function $f(x)$?

- (a) $(0, \infty)$ (b) $(3, \infty)$
(c) $(-\infty, \infty)$ (d) $(-\infty, \infty) \sim \{3\}$

[NDA-2025 (1)]

Consider the following for the two (02) items that follow:

Let $f(x) = (1, 1), (2, 4), (3, 7), (4, 10)$

118. If $f(x) = px + q$, then what is the value of $(p + q)$?

- (a) -1 (b) 0
(c) 1 (d) 5

[NDA-2025 (1)]

119. Consider the following statements:

If is one-one function.

If is onto function if the codomain is the set of natural numbers.

Which of the statements given above is/are correct?

- (a) I only (b) II only
(c) Both I and II (d) neither I nor II

[NDA-2025 (1)]

Consider the following for the two (02) items that follow:

The function $f(x)$ satisfies $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$ for all positive real

values of x and y , and $f(2) = 3$

120. What is $f(16)$ equal to?

- (a) 18 (b) 27
(c) 54 (d) 81

[NDA-2025 (1)]

121. What is $f(1)f(4)$ equal to?

- (a) 4 (b) 8
(c) 9 (d) 18

[NDA-2025 (1)]

Consider the following for the two (02) items that follow:

A function f is such that $f(xy) = f(x + y)$ for all real values of x and y , and $f(5) = 10$

122. What is $f(0)$ equal to?

- (a) 0 (b) 1
(c) 5 (d) 10

[NDA-2025 (1)]

123. What is $f(20) + f(-20)$ equal to?

- (a) 0 (b) 10
(c) 20 (d) 40

[NDA-2025 (1)]

For the following **two (02) items:**

Let $f(x) = ax^2 + bx + c$ be a quadratic polynomial such that $f(1) = f(4) = 2$. Further, 2 is a root of $f(x) = 0$.

124. What is the other root of $f(x) = 0$?

- (a) 1 (b) 2.
(c) 3 (d) Cannot be determined

[NDA-2025 (2)]

125. What is $(a + b + c)$ equal to?

- (a) 0 (b) 1
(c) 2 (d) Cannot be determined

[NDA-2025 (2)]

For the following **two items:**

Consider the function

$$f(x) = \frac{x}{1-x} \quad (x > 0, x \neq 1)$$

126. What is $\frac{f(x)}{f(x+1)}$ equal to?

- (a) $-f(x^2)$ (b) $-f(\sqrt{x})$
(c) $f(x^2)$ (d) $f(x-1)$

[NDA-2025 (2)]

127. What is $(1-x)f(\sqrt{x}) + xf(\sqrt{x}+1)$ equal to?

- (a) $-f(x)$ (b) $f(x)$
(c) x (d) 0

[NDA-2025 (2)]

For the following next **one** items:

Consider the function $f(x) = 1 - \sqrt[3]{(x-1)^2}$

128. What is the domain of the function?

- (a) $(1, \infty)$ (b) $(-\infty, \infty)$
(c) $(0, \infty)$ (d) $(-\infty, \infty) - 1$

[NDA-2025 (2)]

For the following **two** items:

Consider the function $f(x) = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$

129. What is $f \circ f \circ f \circ f \circ f(0)$ equal to?

- (a) 0 (b) 1
(c) 5 (d) 10

[NDA-2025 (2)]

130. What is the inverse of the function

- (a) $\log_{10}(2x-1)$ (b) $\frac{1}{2} \log_{10}(2x-1)$
(c) $\frac{1}{4} \log_{10}\left(\frac{2x}{2-x}\right)$ (d) $\frac{1}{2} \log_{10}\left(\frac{1+x}{1-x}\right)$

[NDA-2025 (2)]

ANSWER KEY

1.	d	2.	a	3.	b	4.	c	5.	c	6.	c	7.	b	8.	c	9.	b	10.	c
11.	d	12.	b	13.	a	14.	c	15.	c	16.	d	17.	c	18.	b	19.	c	20.	b
21.	b	22.	d	23.	a	24.	b	25.	a	26.	a	27.	c	28.	b	29.	a	30.	b
31.	b	32.	a	33.	c	34.	c	35.	b	36.	d	37.	d	38.	c	39.	b	40.	c
41.	a	42.	b	43.	b	44.	d	45.	a	46.	d	47.	a	48.	d	49.	b	50.	d
51.	c	52.	d	53.	d	54.	c	55.	c	56.	c	57.	b	58.	b	59.	d	60.	b
61.	c	62.	a	63.	d	64.	d	65.	c	66.	c	67.	c	68.	b	69.	a	70.	b
71.	c	72.	a	73.	b	74.	b	75.	b	76.	b	77.	d	78.	a	79.	d	80.	d
81.	c	82.	a	83.	a	84.	b	85.	a	86.	d	87.	a	88.	b	89.	a	90.	b
91.	d	92.	d	93.	a	94.	d	95.	b	96.	c	97.	b	98.	c	99.	b	100.	b
101.	b	102.	d	103.	d	104.	a	105.	a	106.	c	107.	c	108.	c	109.	a	110.	b
111.	a	112.	a	113.	c	114.	d	115.	b	116.	b	117.	c	118.	c	119.	a	120.	d
121.	c	122.	d	123.	c	124.	c	125.	c	126.	a	127.	d	128.	b	129.	a	130.	d

Solutions

Sol. 1. (d)

If $f(t) = t^k$

$f(ab) = (ab)^k = a^k b^k$

Sol. 2. (a)

because its derivative is always positive.

Sol. 3. (b)

$$y = \frac{x+5}{3}$$

for inverse function

$$x = \frac{y+5}{3} \Rightarrow y+5 = 3x$$

$$y = 3x - 5$$

$$f^{-1}(x) = 3x - 5$$

Sol. 4. (c)

$$f = g$$

$$x^3 = y^3$$

then $x = y$

I is correct

$y = x$ is always objective.

II is also correct.

Sol. 5. (c)

$$f(x) = x^2$$

is one-one onto function for $x \geq 0$

Sol. 6. (c)

if $x > 0$ then $y = 1$

if $x < 0$ then $y = -1$

there is only two elements in range $\{-1, 1\}$

Sol. 7. (b)

1 is not in range

codomain $\neq$ range

one-one but not onto

Sol. 8. (c)

$$\{1, 2, 3, \dots\} \rightarrow \{2, 4, 6, \dots\}$$

One-one onto

Sol. 9. (b)

$$f(x) = |x|$$

Given that domain is $\mathbb{Z}$ (all integers)

Codomain is $\mathbb{N}$ (non-negative integers)

range of $|x|$ is $[0, \infty)$

if we put $x = 2$ or -2 then we get $y = 2$

so function is many one

codomain = range so function is onto

Sol. 10. (c)

point 3 is not a function because $\{u\}$ gives two images $\{w\}$ and $\{p\}$

point 2 is not a function because x, y don't have any image.

Sol. 11. (d)

with the help of graph of $\sin x$ and $\cos x$, we can conclude that $\sin x$ is increasing in first quadrant and $\cos x$ is decreasing in first quadrant. both given statements are wrong.

Sol. 12. (b)

$$f(x) = 2x + 3$$

Given that domain is $\mathbb{N}$ (natural numbers)

Codomain is $\mathbb{N}$ (natural numbers)

range of $2x + 3$ is $\{5, 7, 9, 11, 13, \dots\}$

function is one one or injective

codomain is not equal to range so function is into or not surjective.

Sol. 13. (a)

$$\frac{f(x)+1}{f(x)-1} + x = \frac{\frac{x-1}{x+1}+1}{\frac{x-1}{x+1}-1} + x = \frac{2x}{-2} + x = 0$$

Sol. 14. (c)

$$f(2x) = \frac{2x-1}{2x+1}$$

$$f(2x) = \frac{2\left(\frac{1+f(x)}{1-f(x)}\right)-1}{2\left(\frac{1+f(x)}{1-f(x)}\right)+1} = \frac{3f(x)+1}{f(x)+3}$$

Sol. 15. (c)

$$f(f(x)) = -1/x$$

Sol. 16. (d)

$$f(g(x)) = g(f(x))$$

$$a(cx+d) + b = c(ax+b) + d$$

$$ad + b = cb + d$$

$$f(d) = g(b)$$

Sol. 17. (c)

x^3 and e^x is one one function so inverse is defined for these functions.

Sol. 18. (b)

Sol. 19. (c)

if $x > 0$ then $y = 1$

if $x < 0$ then $y = -1$

there is only two elements in range

Sol. 20. (b)

in second option there is g odd times in both side.

Sol. 21. (b)

$$|x| - x > 0$$

$$|x| > x$$

this is only possible for negative real numbers.

Sol. 22. (d)

$$\text{If } f(x) = \log x$$

$$\log mn = \log m + \log n$$

Sol. 23. (a)

$$f(x) = x^3$$

$$f(x_1) = f(x_2)$$

$$x_1^3 = x_2^3 \Rightarrow x_1 = x_2$$

function is one one.

Sol. 24. (b)

$$f(x) = 2x-3, g(x) = x^3+5$$

$$f(g(x)) = 2(x^3+5)-3 = 2x^3+7$$

$$y = 2x^3+7$$

$$y-7 = 2x^3$$

$$x = \left(\frac{y-7}{2}\right)^{\frac{1}{3}}$$

$$f \circ g^{-1}(x) = \left(\frac{x-7}{2}\right)^{\frac{1}{3}}$$

Sol. 25. (a)

$$\Rightarrow y = \frac{x^2}{1+x^2} \Rightarrow y + yx^2 = x^2$$

$$\Rightarrow x^2 = \frac{y}{1-y} \Rightarrow x = \sqrt{\frac{y}{1-y}}$$

$$\Rightarrow f^{-1}(x) = \sqrt{\frac{x}{1-x}}$$

domain of inverse function = range of given

function = $[0, 1)$

Sol. 26. (a)

$$\text{check options } f(x) = \ln\left(\frac{1-x}{1+x}\right)$$

satisfies the function

Sol. 27. (c)

$$f(x) = [x], g(x) = |x|$$

$$\text{gof}(x) = |[x]|$$

$$\text{gof}\left(-\frac{5}{3}\right) = \left|\left[-\frac{5}{3}\right]\right| = |-2| = 2$$

$$\text{fog}\left(-\frac{5}{3}\right) = \left[\left[-\frac{5}{3}\right]\right] = \left[\frac{5}{3}\right] = 1$$

$$\text{gof}\left(-\frac{5}{3}\right) - \text{fog}\left(-\frac{5}{3}\right) = 2 - 1 = 1$$

Sol. 28. (b)

$$\text{fof}\left(-\frac{9}{3}\right) = \left[\left[-\frac{5}{3}\right]\right] = [-2] = -2$$

$$\text{gog}(-2) = 2$$

$$\text{fof}\left(-\frac{9}{3}\right) + \text{gog}(-2) = 0$$

Sol. 29. (a)

$$|x| - x > 0$$

$|x| > x$, this is possible only for negative real numbers.

Sol. 30. (b)

$$\Rightarrow \frac{f(a)}{f(a+1)} = \frac{\frac{a}{a-1}}{\frac{a}{a+1}} = \frac{a^2}{a^2-1} = f(a^2)$$

Sol. 31. (b)

$$f(a) = \frac{a-1}{a+1},$$

$$f(2a) = \frac{2a-1}{2a+1} \neq f(a) + 1$$

$$f\left(\frac{1}{a}\right) = \frac{\frac{1}{a}-1}{\frac{1}{a}+1} = \frac{1-a}{1+a} = -f(a)$$

only 2nd statement is correct.

Sol. 32. (a)

$\cos x$ is one-one function in interval

$[0, \pi]$

Sol. 33. (c)

$f(x) = px + q$ and $g(x) = mx + n$. Then

$$f(g(x)) = g(f(x))$$

$$f(mx+n) = g(px+q)$$

$$p(mx+n) + q = m(px+q) + n$$

$$pn+q = mq+n$$

$$f(n) = g(q)$$

Sol. 34. (c)

$f(x) = x^2 - 3$ is an even function

so $(f \circ f \circ f)(x)$ is also an even function

$$(f \circ f \circ f)(-x) = (f \circ f \circ f)(x)$$

$$(f \circ f \circ f)(-1) = (f \circ f \circ f)(-1)$$

statement 1 is true

$$f(x) = x^2 - 3$$

$$\text{fof}(x) = (x^2 - 3)^2 - 3$$

$$\text{fof}(0) = (0 - 3)^2 - 3 = 6$$

$$(f \circ f \circ f)(x) = ((x^2 - 3)^2 - 3)^2 - 3$$

$$(f \circ f \circ f)(0) = ((0 - 3)^2 - 3)^2 - 3 = -2$$

$$(f \circ f \circ f)(-1) = 4 - 3(f \circ f \circ f)(1) = -3(f \circ f \circ f)(-1)$$

$$= -3(-2) = 6 = (f \circ f)(0)$$

statement 2 also correct

Sol. 35. (b)

Sol. 36. (d)

$x^2 - 1$ is an even function

$x + 3/x$ is an odd function

$|x|$ is an even function

$x^2(x - 3)$ is neither even nor odd function.

Sol. 37. (d)

$$f(x) = |x| - x^3$$

$$f(-x) = |x| + x^3$$

$f(x)$ is not equal to $f(-x)$ so function is not an even function.

$f(x)$ is not equal to $-f(-x)$ so function is not an odd function.

Sol. 38. (c)

$$y = \frac{x^2}{1+x^4} \Rightarrow y + yx^4 = x^2$$

$$\Rightarrow yx^4 - x^2 + y = 0$$

$$\Rightarrow x^2 = \frac{1 \pm \sqrt{1-4y^2}}{2y} \Rightarrow x = \sqrt{\frac{1 \pm \sqrt{1-4y^2}}{2y}}$$

$$f^{-1}(x) = \sqrt{\frac{1 \pm \sqrt{1-4x^2}}{2x}}$$

domain of inverse function = range of given

function = $[0, 1/2]$

Sol. 39. (b)

$$\Rightarrow y = 5^{\ln x}$$

$$\Rightarrow \text{for inverse function } \Rightarrow x = 5^{\ln y} \Rightarrow y^{\ln 5}$$

Sol. 40. (c)

$\Rightarrow x/x$ is always 1, but at not $x = 0$, so graph is constant line discontinuous at $x = 0$

Sol. 41. (a)

$$\Rightarrow f(1) + f(2) + f(3) + \dots + f(999) + f(1000)$$

$$\Rightarrow f(1) = \left[\frac{1}{4} + \frac{1}{1000} \right] = 0$$

$$\Rightarrow f(2) = \left[\frac{1}{4} + \frac{2}{1000} \right] = 0$$

$$\Rightarrow f(3) = \left[\frac{1}{4} + \frac{3}{1000} \right] = 0$$

$$\Rightarrow f(750) = \left[\frac{1}{4} + \frac{750}{1000} \right] = 1$$

$$\Rightarrow f(751) = \left[\frac{1}{4} + \frac{751}{1000} \right] = 1$$

$$\Rightarrow f(1000) = \left[\frac{1}{4} + \frac{1000}{1000} \right] = 1$$

$$\Rightarrow f(1) + f(2) + f(3) + \dots + f(999) + f(1000)$$

$$\Rightarrow 0 + 0 + 0 + \dots + 1 + 1 + 1 + \dots + 1 + 1 = 251 \text{ times one} = 251$$

Sol. 42. (b)

$$\text{If } f(x) = \frac{4x + x^4}{1 + 4x^3} \text{ and } g(x) = \ln \left(\frac{1+x}{1-x} \right),$$

$$\Rightarrow g \left(\frac{e-1}{e+1} \right) = \ln \left[\frac{e+1+e-1}{e+1-e+1} \right] = \ln(e) = 1$$

$$\Rightarrow \text{fog} \left(\frac{e-1}{e+1} \right) = f(1) = \frac{4+1}{1+4} = 1$$

Sol. 43. (b)

$$\Rightarrow x^2 - 4x + [x] = 0 \Rightarrow$$

$$\text{for } x \in [0, 1], [x] = 0$$

$$\Rightarrow x^2 - 4x = 0$$

$$\Rightarrow x = 0, 4 \text{ but } 4 \notin [0, 1]$$

$$\text{for } x \in [1, 2], [x] = 1$$

$$\Rightarrow x^2 - 4x + 1 = 0$$

$$\Rightarrow x = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3} \notin [1, 2]$$

so equation has only one solution

Sol. 44. (d)

Sol. 45. (a)

$$|x| - x > 0$$

$|x| > x$, this is possible only for negative real numbers

Sol. 46. (d)

$$\Rightarrow 4 \sin x - 3 \cos x + 1$$

$$\Rightarrow -\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$$

$$\Rightarrow -5 \leq 4 \sin x - 3 \cos x \leq 5$$

$$\Rightarrow -4 \leq 4 \sin x - 3 \cos x + 1 \leq 6$$

range $[-4, 6]$

Sol. 47. (a)

$$\Rightarrow y = \frac{x^2}{1+x^2} \Rightarrow y + yx^2 = x^2$$

$$\Rightarrow x^2 = \frac{y}{1-y} \Rightarrow x = \sqrt{\frac{y}{1-y}}$$

$$\Rightarrow f^{-1}(x) = \sqrt{\frac{x}{1-x}}$$

domain of inverse function = range of given

function = $[0, 1]$

Sol. 48. (d)

$$\text{given, } f(x) = |x+1|$$

by checking the options, we get

$$(a) f(x^2) = |x^2+1|$$

$$[f(x)]^2 = (x+1)^2 \text{ which implies that } f(x^2) \neq f(x)^2$$

$$(b) f(|x|) = ||x|+1|$$

$$|f(x)| = |x+1| \text{ which implies that } f(|x|) \neq |f(x)|$$

$$(c) f(x+y) = |x+y+1|$$

$$f(x) + f(y) = |x+1| + |y+1| \text{ which implies that}$$

$$f(x+y) \neq f(x) + f(y)$$

Sol. 49. (b)

$$\Rightarrow \text{let } f(x) = x^2 \Rightarrow f(x) = 2x \text{ odd function}$$

Sol. 50. (d)

$$\text{for } \sqrt{x-1} \Rightarrow x \geq 1, \text{ and } x \neq 4$$

Sol. 51. (b)

$$g(x) = f(x) + f'(x) + f''(x)$$

$$\text{let } f(x) = x^2 + 1$$

$$f'(x) = 2x + 1$$

$$f''(x) = 2$$

$$g(x) = x^2 + 2x + 1 + 2$$

$$g(x) = (x+1)^2 + 2 > 0$$

Sol. 52. (d)

Two pre images are not possible for a function so it is not a function it is only a relation.

Sol. 53. (d)

$$f(x) = \frac{x-2}{x+2}, x \neq -2$$

$$\Rightarrow y = \frac{x-2}{x+2} \Rightarrow x = \frac{y-2}{y+2}$$

$$\Rightarrow xy + 2x = y - 2 \Rightarrow 2x + 2 = y - xy$$

$$\Rightarrow y = \frac{x-2}{x+2} \Rightarrow x = \frac{y-2}{y+2}$$

Sol. 54. (c)

$$\Rightarrow \frac{A}{P} = \frac{2\pi r}{\pi r^2} = \frac{2}{r}$$

$$\Rightarrow f(1) + f(2) = \frac{2}{1} + \frac{2}{2} = 3$$

Sol. 55. (c)

$$f(x) = 3^{1+x}$$

$$f(x) f(y) f(z) = 3^{1+x} \cdot 3^{1+y} \cdot 3^{1+z}$$

$$3^{1+2+x+y+z} = f(x+y+z+2)$$

Sol. 56. (c)

$$\sqrt{(2-x)(x-3)}$$

$$\Rightarrow (2-x)(x-3) \geq 0$$

$$\Rightarrow (x-2)(x-3) \leq 0$$

$$\Rightarrow x \in [2, 3]$$

Sol. 57. (b)

$$f(x) = \ln(\sqrt{x^2+1} - x)$$

$$f(-x) = \ln(\sqrt{x^2+1} + x)$$

$$f(x) + f(-x) =$$

$$\ln(\sqrt{x^2+1} - x) + \ln(\sqrt{x^2+1} + x)$$

$$= \ln(\sqrt{x^2+1} - x)(\sqrt{x^2+1} + x)$$

$$= \log 1 = 0 \text{ (odd function)}$$

Sol. 58. (b)

$$g[f(x)] = g(x^2 + 2x - 5) = 5(x^2 + 2x - 5) + 30$$

$$5x^2 + 10x - 25 + 30 = 5x^2 + 10x + 5$$

$$5(x^2 + 2x + 1) = 5(x+1)^2 = x = -1, -1$$

Sol. 59. (d)

$$f[g(x)] = f[5x + 30] =$$

$$[(5x + 30)^2 + 2(5x + 30) - 5]$$

polynomial of degree 2

$$g[g(x)] = 5(5x + 30) + 30$$

polynomial of degree 1

Sol. 60. (b)

$$h(x) = 5(x^2 + 2x - 5) - x(5x + 30)$$

$$h(x) = -20x - 25$$

$$h'(x) = -20$$

Sol. 61. (c)

put $x = 0, 1, -1$ in options

Sol. 62. (a)

$$\text{hogofof}(x) = \text{hog}(x^2) = h(\tan x^2) = \ln \tan x^2$$

$$\text{put } x = \frac{\sqrt{\pi}}{2} \text{ then } \text{hogofof} \left(\frac{\sqrt{\pi}}{2} \right) = \ln 1 = 0$$

Sol. 63. (d)

$$\text{fofof}(x) = \text{fof}(x^2) = f(x^4) = x^8$$

$$\text{put } x = 2 \text{ then } \text{fofof}(x) = 2^8 = 256$$

Sol. 64. (d)

$$\text{time period of } \sin ax \text{ is } \frac{2\pi}{a}$$

Sol. 65. (c)

$$\text{time period of } \cos ax \text{ is } \frac{2\pi}{a}$$

Sol. 66. (c)

Time period of $g(x)$ and $h(x)$ is LCM of above LCM of 8π and $5\pi/2$ is 40π

Sol. 67. (c)

$$f(x) = 3x^2 - 5x + p$$

$$f(0) = p$$

$$f(1) = 3 - 5 + p = p - 2$$

$f(0)$ and $f(1)$ is of opposite sign so

$$f(0)f(1) < 1$$

$$p(p-2) < 0$$

$$0 < p < 2$$

Sol. 68. (b)

Domain of $\cos^{-1} x$ is $[-1, 1]$

$$-1 \leq (x-2) \leq 1$$

$$1 \leq x \leq 3$$

Sol. 69. (a)

$$f(x) = 2x - x^2$$

$$f(x+2) + f(x-2) \text{ when } x = 0$$

i.e. $f(2) + f(-2)$
 $= 0 - 8 = -8$

Sol. 70. (b)
 domain = $R - \{1\}$
 Range = $R - \{0\}$

Sol. 71. (c)
 Range of $f(x)$ in 2nd statement is $\{2, 3, 4, 5, \dots\}$
 so function is onto

Sol. 72. (a)

Sol. 73. (b)
 $f(x+1) = x^2 - 3x + 2$
 put $x-1$ at the position of x
 $f(x) = (x-1)^2 - 3(x-1) + 2$
 $f(x) = x^2 - 5x + 6$

Sol. 74. (b)
 Period of $\sin^2 x$ is π
 $f(x) = \log(2 + \sin^2 x)$
 $f(\pi+x) = \log(2 + \sin^2(\pi+x))$
 $= \log(2 + \sin^2 x)$

Sol. 75. (b)
 $f(x) = 1 - \sin x$
 T.f. Range of $f(x)$
 $-1 \leq \sin x \leq 1$
 $-1 \geq -\sin x \geq -1$

Or $-1 \leq 1 - \sin x \leq 1 + 1$
 $0 \leq 1 - \sin x \leq 2$
 $[0, 2]$

Sol. 76. (b)
 statement 1 is wrong because all functions are relations but all relations are not a functions.

Sol. 77. (d)
 $4f(x) - f\left(\frac{1}{x}\right) = 4x^2 - \frac{1}{x^2} \dots (1)$

Replace x with $1/x$

$4f\left(\frac{1}{x}\right) - f(x) = \frac{4}{x^2} - x^2 \dots (2)$

By solving equations $15f(x) = 15x^2$
 $f(x) = x^2 \Rightarrow f(2) = 4$

Sol. 78. (a)
 $f(x) = 4x + 3$
 $\text{fof}(x) = 4(4x+3) + 3$
 $= 16x + 15$
 $\text{fofof}(x) = 4(16x+15) + 3$
 $= 64x + 63$
 $\text{fofof}(-1) = -64 + 63 = -1$

Sol. 79. (d)
 By graph of $y = 10^x$
 all statements are correct

Sol. 80. (d)
 $f(x) = \sqrt{1-x^2} + 2x - 1$
 $= \sqrt{2x-x^2}$
 $\therefore 2x-x^2 \geq 0$
 $x^2 - 2x \leq 0 \Rightarrow x \in [0, 2]$

Sol. 81. (c)
 $f(x)$ = highest prime factor of x
 $f(7) = 7$
 $f(8) = 2$
 $f(9) = 3$
 $f(10) = 5$
 $f(11) = 11$
 $f(12) = 3$
 $f(13) = 13$
 $f(14) = 7$
 $f(15) = 5$
 $f(16) = 2$

Range of $f(x) = \{2, 3, 5, 7, 11, 13\}$
 number of elements in range = 6

Sol. 82. (a)

$z = [y]$ and $y = [x] - x$
 $z = [[x] - x]$
 let $x = Z + f$ (Z is integer and f is decimal part of any number)
 $z = [[Z + f] - (Z + f)]$
 $z = [Z - (Z + f)]$
 $z = [-f]$
 $z = -1$

Sol. 83. (a)
 $f(x) = 4x + 1$ and $g(x) = kx + 2$
 $\text{fog}(x) = \text{gof}(x)$
 $f[g(x)] = g[f(x)]$
 $4(kx+2) + 1 = k(4x+1) + 2$
 $4kx + 8 + 1 = 4kx + k + 2$
 $9 = k + 2$
 $k = 7$

Sol. 84. (b)
 $f(\alpha) = \sqrt{\sec^2 \alpha - 1} = \sqrt{\tan^2 \alpha} = \tan \alpha$
 $\frac{f(\alpha) + f(\beta)}{1 - f(\alpha)f(\beta)} = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \tan(\alpha + \beta)$
 $= \tan(\alpha - \beta) = f(\alpha - \beta)$

Sol. 85. (a)
 $f(x) = \ln(x + \sqrt{1+x^2})$
 $f(-x) = \ln(-x + \sqrt{1+x^2})$
 $f(x) + f(-x)$
 $\ln(x + \sqrt{1+x^2}) + \ln(-x + \sqrt{1+x^2})$
 $\ln(x + \sqrt{1+x^2})(-x + \sqrt{1+x^2})$
 $\ln(1+x^2-x^2) = \log 1 = 0$
 so $f(x)$ is an odd function

Sol. 86.
Sol. 87. (a)
 $f(x) = x^2 + 2$ and $g(x) = 2x - 3$
 $\text{fog}(x) = (2x-3)^2 + 2$
 $\text{fog}(1) = (2-3)^2 + 2 = 3$
Sol. 88. (b)
 $f(x) = x + |x|$
 for $x > 0$, $f(x) = 2x$
 and for $x < 0$, $f(x) = x - x = 0$
 so range is $[0, \infty)$

Sol. 89. (a)
 $f(x) = x(4x^2 - 3)$
 so $f(\sin \theta) = \sin \theta (4\sin^2 \theta - 3)$
 $= - (3\sin \theta - 4\sin^3 \theta) = -\sin 3\theta$

Sol. 90. (b)
 $f(x) = \sin[\pi^2]x + \cos[-\pi^2]x$
 $f(x) = \sin[3.14^2]x + \cos[-3.14^2]x$
 $f(x) = \sin 9x + \cos(-10)x$
 $f(x) = \sin 9x + \cos(10)x$
 $f\left(\frac{\pi}{2}\right) = \sin 9\left(\frac{\pi}{2}\right) + \cos(10)\left(\frac{\pi}{2}\right) = 1 - 1 = 0$

Sol. 91. (d)
 $f(x) = \sin 9x + \cos(10)x$
 $f\left(\frac{\pi}{4}\right) = \sin 9\left(\frac{\pi}{4}\right) + \cos(10)\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} + 0 = \frac{1}{\sqrt{2}}$

Sol. 92. (d)
 $f(x) = \frac{2x+3}{3x+5}$
 if $3x+5 = 0$ then $x = -5/3$ i.e. $-5/3$ is not in domain
 so domain = $R - \{-5/3\}$
 Now by concept of inverse function
 $y = \frac{2x+3}{3x+5}$

replace x with y and y with x

$x = \frac{2y+3}{3y+5}$
 $3xy + 5x = 2y + 3$
 $y(3x-2) = 3-5x$
 $y = \frac{3-5x}{3x-2} = f^{-1}(x)$
 domain of inverse function is $R - \{3/2\}$
 so Range of $f(x)$ is $R - \{3/2\}$

Sol. 93. (a)
 relation f defined by

$f(x) = \begin{cases} x^3, & 0 \leq x \leq 2 \\ 4x, & 2 \leq x \leq 8 \end{cases}$

$f(2) = 8$, by both above equations.
 so f is a function
 relation g defined by

$g(x) = \begin{cases} x^2, & 0 \leq x \leq 4 \\ 3x, & 4 \leq x \leq 8 \end{cases}$ is a function

$f(4) = 16$ by first equation
 $f(4) = 12$ by second equation
 but functions states that two images of a single element is not possible.
 so g is not a function

Sol. 94. (d)
 $f(1) = 0.5$ given

$f(x-y) = \frac{f(x)}{f(y)}$

$f(1) = 0.5$ given
 put $x=2$ and $y=1$

$f(2-1) = f(1) = \frac{f(2)}{f(1)}$

$f(2) = f(1).f(1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

put $x=3$ and $y=1$

$f(3-1) = f(2) = \frac{f(3)}{f(1)}$

$f(3) = f(2).f(1) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$

by above pattern

we can conclude that $f(x) = \frac{1}{2^x}$

$f(2) + f(3) + f(4) + f(5) + f(6)$
 $= \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} = \frac{31}{64}$

Sol. 95. (b)

$f(x) = x^2 - 1$ and $\text{gof}(x) = x - \sqrt{x} + 1$

let $f(x) = y$

$y = x^2 - 1$ so $x = \sqrt{1+y}$

$\text{gof}(x) = x - \sqrt{x} + 1$

if $f(x) = y$ then

$g(y) = x - \sqrt{x} + 1 = \sqrt{1+y} - \sqrt{\sqrt{1+y}} + 1$

$g(y) = \sqrt{1+y} - \sqrt[4]{1+y} + 1$

replace y with x

$g(x) = \sqrt{1+x} - \sqrt[4]{1+x} + 1$

Sol. 96. (c)

$g(x) = \sqrt{1+x} - \sqrt[4]{1+x} + 1$

$g(15) = \sqrt{16} - \sqrt[4]{16} + 1 = 4 - 2 + 1 = 3$

Sol. 97. (b)

$2f(x) + f\left(\frac{1}{x}\right) = x + 3 \dots \dots \dots (i)$

replace x with $1/x$

$$2f\left(\frac{1}{x}\right) + f(x) = \frac{1}{x} + 3 \quad \dots\dots\dots(ii)$$

by solving above equations

$$f(x) = \frac{2x}{3} - \frac{1}{3x} + 1$$

$$f\left(\frac{1}{2}\right) = \frac{2}{3}\left(\frac{1}{2}\right) - \frac{1}{3}\left(\frac{2}{1}\right) + 1 = \frac{2}{3}$$

Sol. 98. (c)

by above solution

$$f(x) = \frac{2x}{3} - \frac{1}{3x} + 1$$

$$f'(x) = \frac{2}{3} + \frac{1}{3x^2}$$

$$f'\left(\frac{1}{2}\right) = \frac{2}{3} + \frac{4}{3} = 2$$

Sol. 99. (b)

$$f(x) = \pi + \sin^2 x$$

$$0 \leq \sin^2 x \leq 1$$

$$\pi \leq \pi + \sin^2 x \leq \pi + 1$$

so range of function will be $[\pi, \pi + 1]$

Sol. 100. (b)

$$f(x) = \pi + \sin^2 x$$

$$f(\pi + x) = \pi + \sin^2(\pi + x) = \pi + \sin^2 x$$

$$f(\pi + x) = f(x)$$

so time period is π

Sol. 101. (b)

$$f(x) = \sqrt{2-x} + \sqrt{2+x}$$

$$2-x \geq 0 \text{ so } x \leq 2$$

$$\text{and } 2+x \geq 0 \text{ so } x \geq -2$$

$$\text{so } -2 \leq x \leq 2$$

Sol. 102. (d)

$$3f(x) + f\left(\frac{1}{x}\right) = \frac{1}{x} + 1 \quad \dots\dots\dots(i)$$

replace x with 1/x

$$3f\left(\frac{1}{x}\right) + f(x) = x + 1 \quad \dots\dots\dots(ii)$$

by solving above equations

$$(x) = \frac{3}{8x} - \frac{x}{8} + \frac{1}{4}$$

Sol. 103. (d)

$$f(x) = ax - b \text{ and } g(x) = cx + d$$

$$fog(x) = gof(x)$$

$$f[g(x)] = g[f(x)]$$

$$a(cx + d) - b = c(ax - b) + d$$

$$acx + ad - b = acx - bc + d$$

$$ad - b = -bc + d$$

$$f(d) = -bc + d$$

$$f(d) + bc + d = 2d$$

$$f(d) + g(b) = 2d$$

Sol. 104. (a)

$$f(x) = \frac{1}{\sqrt{|x| - x}}$$

$|x| > x$ i.e. x should be negative

$$g(x) = \frac{1}{\sqrt{x - |x|}}$$

$x > |x|$ that is not possible

so $f(x)$ has some domain but $g(x)$ has no domain.

Sol. 105. (a)

$$f(x) = x|x|$$

$$\text{for } 0 \leq x < 1, f(x) = x^2$$

$$\text{for } -1 < x < 0, f(x) = -x^2$$

for unique value of x from -1 to 1, $f(x)$ gives unique value.

so function is one one

from -1 to 1 range of $f(x)$ is -1 to 1, so function is onto.

$$g(x) = \cos(\pi x)$$

for two different values of x from -1 to 1, $g(x)$

gives one common value.

$$\text{for example put } x = \frac{1}{2} \text{ or } -\frac{1}{2}, g(x) = 0$$

so function is many one

from -1 to 1 range of $g(x)$ is -1 to 1, so function is onto.

one one onto functions are called bijective function.

statement 1 is correct and 2 is incorrect.

Sol. 106. (c)

if there is n elements in A and 2 elements in B then number of onto functions = $2^n - 2 = 2^5 - 2 = 30$

Sol. 107. (c)

$$f(t) = \ln(t + \sqrt{1+t^2})$$

$$f(-t) = \ln(-t + \sqrt{1+t^2})$$

$$f(t) + f(-t) = 0$$

so $f(t)$ is an odd function

$$g(t) = \tan(f(t))$$

$$g(-t) = \tan(f(-t)) = \tan(-f(t))$$

$$= -\tan(f(t)) = -g(t)$$

so $g(t)$ is an odd function

Sol. 108. (c)

$$fog(x) = \cos^2 \sqrt{x}$$

$$gof(x) = \cos x$$

by inspection of both composite functions we can conclude that $f(x) = \cos^2 x$ and $g(x) = \sqrt{x}$

Sol. 109. (a)

by above answer $g(x) = \sqrt{x}$

Sol. 110. (b)

$$f(x) = [x]^2 - [x^2]$$

let $0.001 = h$ (very less value)

$$f(0.999) + f(1.001)$$

$$f(1 - 0.001) + f(1 + 0.001)$$

$$f(1 - h) + f(1 + h)$$

$$[1 - h]^2 - [(1 - h)^2] + [1 + h]^2 - [(1 + h)^2]$$

$$0 - 0 + 1 - 1 = 0$$

Sol. 111. (a)

$$z = [y] \text{ and } y = [x] - x$$

$$z = [[x] - x]$$

let $x = Z + f$ (Z is integer and f is decimal part of any number)

$$z = [[Z + f] - (Z + f)]$$

$$z = [Z - (Z + f)]$$

$$z = [-f]$$

$$z = -1$$

Sol. 112. (a)

$$f(x) = 4x + 1 \text{ and } g(x) = kx + 2$$

$$fog(x) = gof(x)$$

$$f[g(x)] = g[f(x)]$$

$$4(kx + 2) + 1 = k(4x + 1) + 2$$

$$4kx + 8 + 1 = 4kx + k + 2$$

$$9 = k + 2$$

$$k = 7$$

Sol. 113. (c)

$$f(x) = 4x^2 + 1$$

$f(2x)$ is mean of $f(x)$ and $f(4x)$

$$f(2x) = \sqrt{f(x)f(4x)}$$

$$4(2x)^2 + 1 = (4x^2 + 1)(4(4x)^2 + 1)$$

$$(16x^2 + 1)^2 = (4x^2 + 1)(64x^2 + 1)$$

$$256x^4 + 32x^2 + 1 = 256x^4 + 68x^2 + 1$$

$$36x^2 = 0$$

$$x = 0$$

x has only one real value

Sol. 114. (d)

$$[x]^2 - 30[x] + 221 = 0$$

$$([x] - 17)([x] - 13) = 0$$

$$[x] = 17 \text{ or } [x] = 13$$

if $[x] = 17$ then its integer solution is $x = 17$

$[x] = 13$ then its integer solution is $x = 13$

so sum of both possible values = 30

Sol. 115. (a)

$$f(x)f(y) = f(xy)$$

$$\text{i.e. } f(x) = x^n$$

$$\text{put } x = 2$$

$$f(2) = (2)^n = 4 \text{ so } n = 2$$

$$\text{then } f(x) = x^2$$

$$f(1/2) = 1/4$$

Sol. 116. (b)

$$y = \frac{1}{(1 - \cos x)}$$

We know that

$$-1 \leq \cos x \leq 1$$

$$0 \leq 1 - \cos x \leq 2$$

Reciprocal both sides

$$\infty > \frac{1}{(1 - \cos x)} \geq 0.5$$

Sol. 117. (c)

Domain of $|x - 3|$ is $(-\infty, \infty)$

Sol. 118. (c)

$$f(x) = px + q$$

$$\text{put } x = 1$$

$$f(1) = p + q$$

$$1 = p + q$$

Sol. 119. (a)

$$f(x) = px + q$$

$$\text{put } x = 1$$

$$f(1) = p + q$$

$$1 = p + q \quad \dots\dots\dots(i)$$

$$\text{Put } x = 2$$

$$f(2) = 2p + q$$

$$4 = 2p + q \quad \dots\dots\dots(ii)$$

By above 2 equations $p = 3$ and $q = -1$

$$\text{So } f(x) = 3x - 2$$

Function is linear function, its graph is straight line, so function is one one.

There are only 4 elements in range $\{1, 4, 7, 10\}$

Codomain is not equal to range

So function is into function.

Sol. 120. (d)

$$f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$$

Given that $f(2) = 3$

$$\text{Put } x = y = 2$$

$$f\left(\frac{2}{2}\right) = \frac{f(2)}{f(2)} \Rightarrow f(1) = 1$$

$$\text{put } x = 4, y = 2$$

$$f\left(\frac{4}{2}\right) = \frac{f(4)}{f(2)} \Rightarrow f(2) = \frac{f(4)}{3} = 3$$

$$f(4) = 9$$

$$\text{put } x = 16, y = 4$$

$$f\left(\frac{16}{4}\right) = \frac{f(16)}{f(4)} \Rightarrow f(4) = \frac{f(16)}{9} = 9$$

$$f(16) = 81$$

Sol. 121. (c)

by above solution

$$f(1).f(4) = (1)(9) = 9$$

Sol. 122. (d)

$$f(xy) = f(x + y)$$

above expression is satisfies for constant function.

$$f(0) = f(5) = 10$$

Sol. 123. (c)

$$f(xy) = f(x + y)$$

above expression is satisfies for constant function.

$$f(20) = f(-20) = f(5) = 10$$

$$\text{so } f(20) + f(-20) = 10 + 10 = 20$$

Sol. 124. (c)

$$f(x) = ax^2 + bx + c$$

$$f(1) = f(4) = 2$$

$$a + b + c = 2 \quad \dots(i)$$

$$16a + 4b + c = 2 \quad \dots(ii)$$

$$2 \text{ is root of } f(x) = 0$$

$$4a + 2b + c = 0 \quad \dots(iii)$$

By above 3 equations $a = 1$, $b = -5$, $c = 6$

$$\text{So } f(x) = x^2 - 5x + 6 = 0$$

$x = 2, 3$ are roots

one root 2 is given in equation so other root is 3

Sol. 125. (c)

by above solution $a + b + c = 1 - 5 + 6 = 2$

Sol. 126. (a)

$$f(x) = \frac{x}{1-x}$$

$$\frac{f(x)}{f(x+1)} = \frac{\frac{x}{1-x}}{\frac{x+1}{1-x-1}} = -\frac{x^2}{1-x^2} = -f(x^2)$$

Sol. 127. (d)

$$(1-x)f(\sqrt{x}) + xf(\sqrt{x}+1)$$

$$(1-x)\frac{\sqrt{x}}{1-\sqrt{x}} + x\frac{(\sqrt{x}+1)}{1-\sqrt{x}-1}$$

$$= (1+\sqrt{x})\sqrt{x} - \sqrt{x}(\sqrt{x}+1) = 0$$

Sol. 128. (b)

$$f(x) = 1 - (x-1)^{\frac{2}{3}}$$

We can put any value of x

So domain is $(-\infty, \infty)$

Sol. 129. (a)

$$f(0) = \frac{10^0 - 10^0}{10^0 + 10^0} = 0$$

$$f \circ f(x) = \frac{10^{\left(\frac{10^x - 10^{-x}}{10^x + 10^{-x}}\right)} - 10^{\left(\frac{10^x - 10^{-x}}{10^x + 10^{-x}}\right)}}{10^{\left(\frac{10^x - 10^{-x}}{10^x + 10^{-x}}\right)} + 10^{\left(\frac{10^x - 10^{-x}}{10^x + 10^{-x}}\right)}}$$

$$f \circ f(0) = 0$$

so on $f \circ f \circ f \circ f \circ f(0) = 0$

Sol. 130. (d)

$$f(x) = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$$

Componendo and dividend

$$\frac{f(x)-1}{f(x)+1} = -\frac{10^x}{10^{-x}} = -10^{2x}$$

For inverse function

$$\frac{x-1}{x+1} = -10^{2y}$$

$$y = \frac{1}{2} \log_{10} \left(\frac{1+x}{1-x} \right)$$